Amdahl's Law

Amdahl's law was proposed by Gene Amdahl in 1967. It provides an expression to find the maximum expected improvement (or enhancement) to an overall system when only part of the system is improved. Amdahl's law is mostly used in parallel computing to account for the theoretical maximum speedup using multiple processors.

The statement of the Amdahl's law is:
"the performance improvement, to be gained from using some faster mode of execution, is limited by the fraction of the time the faster mode can be used".

Amdahl's law deals with the potential Speedup of a program using, for example, multiple processors compared to a single processor.

Speedup is the ratio that tells *how much faster a task will run using the computer with the enhancement as opposed to the original computer(without using that enhancement).*

Performance of a system that has been sped up can be defined as:

$$\boldsymbol{Speedup} = \frac{\boldsymbol{Execution\ time\ before\ improvement\ (or\ enhancement)}}{\boldsymbol{Execution\ time\ after\ improvement\ (or\ enhancement)}}$$

or

$$\boldsymbol{Speedup} = \frac{\boldsymbol{Execution\ time\ without\ improvement\ (or\ enhancement)}}{\boldsymbol{Execution\ time\ with\ improvement\ (or\ enhancement)}}$$

Consider a program running on a single processor such that *a fraction of the execution time involves the code that is inherently sequential*, and *a fraction that involves the code that is infinitely parallelizable with no scheduling overhead*. Let **T** be the total execution time of the program using a single processor. Then, using Amdahl's law, we can find the speedup for the same program running on a parallel processing system with **N** Processors that fully exploits the parallel portion of the program.

Here, the statement "*a fraction of the execution time involves code that is inherently sequential"* can be considered as normal process i.e., fraction of the program without any improvement (or enhancement).
Whereas, the statement "*a fraction f that involves code that is infinitely parallelizable with no scheduling overhead"* informs that it is the fraction of the program with improvement (or enhancement) i.e., fraction enhanced (or improved).
Here, the total execution time of the program will be *the sum of the execution time for the sequential code and the execution time for the parallel code*.

Let Speedup be denoted by "S", and fraction enhanced (or improved) be denoted by "f_E", then the fraction of the program without enhancement (or improvement) will be denoted by "$1 - f_E$".

We can say that

- Execution time of the program using single processor (without improvement or enhancement) equals $\mathbf{T(1-f_E)+\ T\ (f_E)}$
- Execution time of the program using N processors (with improvement or enhancement) equals $\mathbf{T(1-f_E)+\ T\ (\frac{f_E}{N})}$ *since parallel code will be divided among the N processors.*

According to Amdahl's law:

$$\boldsymbol{Speedup = \frac{Execution\ time\ without\ improvement\ (or\ enhancement)}{Execution\ time\ with\ improvement\ (or\ enhancement)}}$$

$$\boldsymbol{S = \frac{\mathbf{T}(1-f_E)+\ \mathbf{T}\ (f_E)}{\mathbf{T}(1-f_E)+\ \mathbf{T}\ (\frac{f_E}{N})}}$$

$$\boldsymbol{S = \frac{\mathbf{T}(1-f_E+f_E)}{\mathbf{T}\{(1-f_E)+\left(\frac{f_E}{N}\right)\}}}$$

$$\boldsymbol{S = \frac{1}{(1-f_E)+\left(\frac{f_E}{N}\right)}}$$

We can write,

$$\boldsymbol{Speedup = \frac{1}{(1-fraction\ enhanced)+\left(\frac{fraction\ enhanced}{factor\ of\ enhancement}\right)}}$$

Let the factor of enhancement (or improvement) be denoted by "K", then the above equation can be written as:

$$\boldsymbol{S = \frac{1}{\left((1-\ f_E\ )+\left(\frac{f_E}{K}\right)\right)}}$$

or

$$\boldsymbol{S = \left((1-\ f_E\ )+\left(\frac{f_E}{K}\right)\right)^{-1}}$$

The problem types which can be solved using Amdahl's law are as follows:

- Determine S given f_E and K
- Determine K given S and f_E
- Determine f_E given S and K